

Statistics Calculator Guide

How to Use the MDragon Statistics Calculator

Overview

The Statistics Calculator provides comprehensive descriptive statistics and visualizations for your data.

Step-by-Step Instructions

Step 1: Access the Calculator

Navigate to: **Calculators** → **Statistics**

Step 2: Enter Your Data

Option A - Manual Entry:

1. Type numbers separated by commas or spaces
2. Example:

Option B - Paste Data:

1. Copy data from spreadsheet
2. Paste into input field
3. Calculator automatically detects format

Step 3: Calculate Results

Click "**Calculate**" button to see:

Measures of Central Tendency

- **Mean ($\bar{x}$):** Average value

- **Median:** Middle value when sorted
- **Mode:** Most frequent value

Measures of Spread

- **Range:** Max - Min
- **Variance (s^2):** Average squared deviation
- **Standard Deviation (s):** Square root of variance
- **IQR:** Q3 - Q1 (middle 50%)

Quartiles

- **Q1:** 25th percentile
- **Q2:** Median (50th percentile)
- **Q3:** 75th percentile

Example Usage

Data: Test scores: 65, 70, 75, 80, 85, 90, 95

Results:

- Mean: 80
- Median: 80
- Std Dev: 10.8
- Range: 30

Interpretation:

- Average score is 80
- Half scored above/below 80
- Typical deviation from mean is ~11 points

Tips

- ✓ Remove outliers if appropriate
 - ✓ Use at least 5-10 data points
 - ✓ Check for data entry errors
 - ✓ Compare mean vs median for skewness
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MDragon Data Tools | Statistics Education Platform

mondragon-developer.github.io/statools